HDOC Day-49

C Programming Activity - Pivot Integer Using Pointers

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Problem Statement

Write a C program that reads a positive integer **n**. Pass **n** to a function named **pivot_integer()** using its address. Use pointers to pass the value between the calling function and **pivot_integer()**. The function must find a pivot integer **x** between 1 and n such that the sum from 1 to x is equal to the sum from x to n. Display x if it exists; otherwise display -1.

C Program

```
#include <stdio.h>

void pivot_integer(int *n)
{
    int x, i;
    int leftSum, rightSum;

    for (x = 1; x <= *n; x++)
    {
        leftSum = 0;
        rightSum = 0;

        for (i = 1; i <= x; i++)
            leftSum += i;

        for (i = x; i <= *n; i++)
            rightSum += i;

        if (leftSum == rightSum)
        {
            printf("Pivot integer: %d\n", x);
            return;
        }
    }

    printf("-1\n");
}

int main()
{
    int n;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    pivot_integer(&n);

    return 0;
}
```

Explanation

The address of **n** is passed from **main()** using **pivot_integer(&n)**. Inside the function, ***n** accesses the original value. For every possible x, the program calculates the sum from 1 to x and from x to n. If both

sums are equal, x is printed as the pivot integer. If no value satisfies the condition, the program prints -1.

Sample Output

Example 1

```
Enter a positive integer: 8
Pivot integer: 6
```

Example 2

```
Enter a positive integer: 1
Pivot integer: 1
```

Example 3

```
Enter a positive integer: 4
-1
```

Result: The program successfully finds the pivot integer using a pointer-based function call and displays -1 when no pivot integer exists.